D.c. extensions from finite-codimensional polyhedral cylinders

Partial answer to Remark 2.4(a) of arXiv:0810.1433

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Status. `candidate_partial_likely_valid`.

Abstract

Veselý and Zajíček ask whether a power-type-2 uniformly convex renorming assumption can be removed or weakened in their d.c.-extension theorem. We remove all renorming assumptions for finite-codimensional polyhedral cylinders. If $T : X \to E$ is onto a finite-dimensional space, $K \subset E$ is a full-dimensional closed convex polyhedron, and $C = T^{-1}(K)$, then every d.c. mapping $F : C \to Y$ for which F and one control function are Lipschitz on bounded parts of C has a d.c. extension to X. The proof replaces the source's radial retraction by a finite-rank piecewise-affine retraction. The ambient normed space X and the target Y are otherwise arbitrary.

1 Source problem

For a convex set D in a normed space, a mapping $G : D \to Y$ is d.c. when there is a continuous convex $g : D \to \mathbb{R}$ such that $y^* \circ G + g$ is convex for every $y^* \in Y^*$ with $\|y^*\| \le 1$. Such a g is a *control function.*

Theorem 2.3 of Veselý–Zajíček [1] gives equivalent extension criteria under the assumption that X admits an equivalent norm with modulus of convexity of power type 2. Remark 2.4(a), reproduced in Figure 1, asks whether that assumption can be omitted or essentially weakened.

The source uses this geometry to prove that the radial retraction onto $\overline{C}$ is d.c.; its remaining composition argument is geometric-free. The following result supplies a different retraction.

2 Idea of the proof

A finite-dimensional polyhedron has a nearest-point projection which is piecewise affine on finitely many polyhedral regions. Finite piecewise-affine maps are d.c. with global Lipschitz controls. If the polyhedron is observed through a finite-dimensional quotient $T : X \to E$, a linear right inverse lifts that projection to a d.c. retraction of the arbitrary normed space X onto $T^{-1}(K)$. The retraction sends bounded sets to bounded sets, so the source's local composition lemma applies ball by ball.

3 The piecewise-affine retraction

We use the following standard finite-dimensional fact in a form that records the control needed below.

Lemma 1 (Finite piecewise-affine maps)**.** *Let E and Z be finite-dimensional normed spaces. Every continuous map $H : E \to Z$ which is affine on each cell of a finite polyhedral subdivision is globally Lipschitz and d.c. with a globally Lipschitz control function.*

Proof. It suffices to treat a scalar component h. Let $\lambda_1 = 0, \ldots, \lambda_N = 0$ be the finite collection of affine hyperplanes containing the facets of the subdivision. Across a common facet, continuity forces the jump of the affine gradient of h to be a scalar multiple of the normal gradient of the corresponding λ_i. For a sufficiently large M,

$$g = M \sum_{i=1}^{N} |\lambda_i|$$

has a normal gradient jump which dominates, with either sign, every such jump of h. The standard facet criterion for convex piecewise-affine functions then shows that $g + h$ and $g - h$ are convex. Thus g controls h, and both h and g are globally Lipschitz.

Choose coordinates on Z, apply the scalar argument to every component, and take a sufficiently large positive sum of the scalar controls. Writing a coefficient a with $|a| \le c$ as a positive combination of the controls for h and $-h$ shows that this single sum controls every member of the dual unit ball of Z. □

Lemma 2 (Polyhedral-cylinder retraction)**.** *Let X be a normed space, let E be finite-dimensional, and let $T : X \to E$ be a continuous linear surjection. If $K \subset E$ is a nonempty closed convex polyhedron, then $C = T^{-1}(K)$ admits a globally Lipschitz d.c. retraction $R : X \to C$ with a globally Lipschitz control. In particular, R maps bounded sets to bounded sets.*

Proof. Equip E with an auxiliary Euclidean norm and let $\pi_K : E \to K$ be nearest-point projection. It is 1-Lipschitz. For each face G of K, the points whose projection lies in the relative interior of G form a polyhedral region, and on that region π_K is orthogonal projection onto the affine hull of G. Since K has finitely many faces, π_K is a finite continuous piecewise-affine map. It is d.c. with a Lipschitz control by Lemma 1.

Because E is finite-dimensional, T has a continuous linear right inverse $S : E \to X$. Define

$$R(x) = x + S\big(\pi_K(Tx) - Tx\big). \tag{1}$$

Linear pre- and postcomposition and addition of an affine map preserve d.c. mappings and Lipschitz controls, so R has the asserted regularity. Also

$$TR(x) = \pi_K(Tx) \in K,$$

while $R(x) = x$ whenever $Tx \in K$. Thus R is a retraction onto C. Its global Lipschitz bound implies bounded-set preservation. □

4 Extension theorem without ambient renorming

For the next statement, "Lipschitz on bounded parts of C" means Lipschitz on $C \cap rB_X$ for every $r > 0$.

Theorem 1 (Polyhedral extension)**.** *Let X, Y be arbitrary normed spaces, E finite-dimensional, $T : X \to E$ a continuous linear surjection, and $K \subset E$ a full-dimensional closed convex polyhedron. Put $C = T^{-1}(K)$. Suppose $F : C \to Y$ is d.c. and has a control function $f : C \to \mathbb{R}$ such that both F and f are Lipschitz on bounded parts of C. Then F admits a d.c. extension to all of X. No renorming hypothesis on X is required.*

Proof. Let $R : X \to C$ be the retraction from Lemma 2 and set

$$\widehat{F} = F \circ R.$$

This extends F. To prove it is d.c., use Lemma 1.12 of the source. Let $A_n = n \operatorname{int} B_X$. Since R is globally Lipschitz, choose $M_n < \infty$ such that

$$R(A_n) \subset B_n := C \cap M_n B_X.$$

Each B_n is convex. By hypothesis, $F|_{B_n}$ is Lipschitz and d.c. with the Lipschitz control $f|_{B_n}$. The map R is d.c., the open convex sets A_n increase to X, and $R(A_n) \subset B_n$. Every hypothesis of the source's composition lemma is therefore satisfied, so $F \circ R$ is d.c. on X. □

Corollary 1. *The conclusion holds, on every normed space, when C is a halfspace, a finite-codimensional slab or box, or more generally the inverse image of any finite-dimensional closed convex polyhedron under a quotient map.*

5 Scope and obstruction to a full upgrade

Theorem 1 gives a genuine class of infinite-dimensional spaces and domains for which the source's renorming assumption disappears. It does strengthen the local hypothesis by requiring F and one control to be Lipschitz on bounded parts of C. It does not settle Theorem 2.3 for an arbitrary convex C.

The remaining difficulty is not the final composition step but the existence of a suitable d.c. retraction. For a general C, the source's radial map multiplies a scalar d.c. function by the full infinite-dimensional vector x. Flat directions of arbitrary norms prevent the obvious radial convex controls from dominating the mixed bilinear curvature. Eight focused routes, including scalar splitting, UMD geometry, and counterexample reductions, are recorded in the accompanying attempt log.

Bounded arXiv searches through 12 August 2026 found no later resolution of Remark 2.4(a) and no statement of Theorem 1. Novelty confidence is moderate-low because the retraction argument is elementary and may be unindexed folklore. Expert review should focus on Lemma 1 and the scope comparison with the source theorem.

References

[1] L. Veselý and L. Zajíček, *On extensions of d.c. functions and convex functions*, arXiv:0810.1433; Journal of Convex Analysis 17 (2010), 427–440.

given by

$$P(x) = \begin{cases} x & \text{if } x \in \overline{C}; \\ \frac{x}{\mu(x)} & \text{if } x \in X \setminus \overline{C}. \end{cases}$$

The function $x \mapsto \max\{1, \mu(x)\}$ is convex and Lipschitz, and its values belong to $[1, \infty)$. The function $t \mapsto \frac{1}{t}$ is convex and Lipschitz on $[1, \infty)$; consequently, by Proposition 1.11, the composed function $x \mapsto \frac{1}{\max\{1,\mu(x)\}}$ is d.c. on X. Moreover, the mapping $B\colon \mathbb{R} \times X \to X$, given by

$$B(t, x) = tx\,,$$

is a continuous bilinear mapping. Since $P(x) = B\left(\frac{1}{\max\{1,\mu(x)\}}, x\right)$, $x \in X$, Proposition 1.13 implies that P is d.c. on X.

Let us show that $\hat{F} := F^* \circ (P|_A)$ is a d.c. extension of F to A. The fact that $\hat{F}$ extends F is obvious. To prove that $\hat{F}$ is d.c., it is sufficient to apply Lemma 1.12 with $B := \overline{C} \cap A$, $\Phi := P|_A$, $\Psi := F^*$ (and A, A_n, B_n as above). Indeed, the assumptions of that lemma are satisfied since $\Phi(A_n) = P(A_n) \subset \overline{C} \cap A_n = B_n$ (note that $P(A_n) \subset A_n$ because $0 \in A_1$) and (3) holds. $\square$

Remark 2.4. (a) We do not know whether the renorming assumption on X in Theorem 2.3 can be omitted or essentially weakened.

(b) The condition (ii) in Theorem 2.3 can be substituted by the following formally weaker condition:

(ii') *some control function of F can be extended to a d.c. function on A.*

Indeed, if f_1 and f_2 are continuous convex functions on A such that $f_1 - f_2$ controls F on C, then also the sum $f_1 + f_2$ controls F on C.

Figure 1: Source page 7: the radial-retraction step and Remark 2.4(a).